

Smart-LMS Navigation Bar & Interface Feature Manual

Detailed User & Developer Specification for All 14 Sidebar Links, Top Navigation Controls & UI Actions

COMPLETE INTERFACE BLUEPRINT • VERSION 1.0.0

1. Overview of the Layout Architecture

The Smart-LMS user interface employs a responsive dual-navigation shell consisting of a collapsible **Sidebar Navigation Bar** on the left and a sticky **Top Navbar** on the top.

- Sidebar Component** (`Sidebar.tsx`): Manages primary navigation routing across all 14 core platform modules. Supports collapsible states (256px expanded $\rightarrow$ 68px collapsed), animated active indicators, and mobile drawer transitions.
- Navbar Component** (`Navbar.tsx`): Manages global search queries, theme switching (Light/Dark mode), notification triggers, user profile summaries, target exam selection, and authentication logout flows.

2. Detailed Explanation of All 14 Sidebar Navigation Links

1. 🏠 Dashboard

Path: /

The central command center for the student. Provides an instant visual summary of daily study missions, current study streak, overall accuracy, readiness score, weak subject alerts, and quick action shortcuts.

Icon: `LayoutDashboard` • Backend Service: `/api/dashboard` • Key Utility: Daily target tracking & streak monitoring.

2. 🤖 AI Mentor

Path: /ai-tutor

An interactive 24/7 AI-powered academic tutor. Allows students to type or paste complex questions, select explanation depth (step-by-step, hint, or complete solution), and receive formatted markdown answers with code syntax highlighting.

Icon: `Bot` • Backend Service: `/api/ai/solve` (Groq LLaMA 3.1 LLM) • Key Utility: On-demand problem solving & instant clarification.

3. 📖 Learning Hub

Path: /study

An interactive document study workspace. Enables users to select uploaded study materials or notes, process text, view automated summaries, and practice with AI-generated flashcard carousels.

Icon: `BookOpen` • Backend Service: `/api/notes` & `/api/books` • Key Utility: Deep reading, flashcard revision, and active recall.

4. 📝 Practice Arena

Path: /quiz

The testing & assessment engine. Generates customized multiple-choice practice quizzes based on chosen subjects, difficulty levels, or custom text. Features countdown exam timers and detailed answer explanation modals.

Icon: `ClipboardList` • Backend Service: `/api/mocktest` • Key Utility: Exam preparation, speed testing, and score accuracy tracking.

5. 📅 Study Planner

Path: /planner

An adaptive academic schedule planner. Automatically balances daily study tasks based on upcoming exam dates, current academic risk scores, and weak subject areas. Includes AI Weekly Strategy Reviews.

Icon: `Calendar` • Backend Service: `/api/planner` • Key Utility: Smart schedule creation & Risk Score reduction.

6. 📊 Progress

Path: /analytics

A comprehensive data visualization dashboard powered by Recharts. Displays weekly study hours, questions attempted, notes created, subject mastery radar charts, and trend metrics.

Icon: [BarChart3](#) • Backend Service: [/api/analytics](#) • Key Utility: Visual performance auditing & readiness evaluation.

7. 📖 Library

Path: /books

A digital repository of reference books, textbooks, and syllabus guides categorized by subject and target exam (JEE, NEET, UPSC, GATE, etc.).

Icon: [Library](#) • Backend Service: [/api/books](#) • Key Utility: Coursework reference & digital book access.

8. 🎥 Learning Videos

Path: /youtube

An educational video discovery and summarization module. Allows students to search YouTube educational videos, view video duration/views, save favorites, and generate one-click AI video summaries.

Icon: [Youtube](#) • Backend Service: [/api/youtube](#) • Key Utility: Visual video learning & AI summary extraction.

9. 📝 Notes

Path: /notes

A full-featured digital notebook for creating, editing, categorizing, and searching study notes. Supports Markdown formatting, tagging, and AI note refinement.

Icon: [FileText](#) • Backend Service: [/api/notes](#) • Key Utility: Personal knowledge management & revision notes.

10. ⌚ Focus Timer

Path: /focus

A dedicated Pomodoro timer built to maximize study concentration. Features customizable work/break cycles, ambient focus audio, study session logs, and streak counters.

Icon: [Timer](#) • Backend Service: Local Component State + User XP • Key Utility: Time management & deep work enforcement.

11. 🧠 Critical Thinking

Path: /critical

A gamified cognitive brain-training zone featuring logical reasoning puzzles, math brain games, and strategic scenarios. Tracks Intelligence Scores and awards badges like "Focus Beast".

Icon: [Brain](#) • Backend Service: [/api/brain](#) • Key Utility: Cognitive training & XP/Level ranking.

12. 📁 Upload Material

Path: /upload

The file ingestion portal. Enables drag-and-drop file uploading for PDFs, handwritten note photos, and study slides. Automatically triggers Tesseract OCR text extraction.

Icon: [Upload](#) • Backend Service: [/api/upload](#) & [/api/ocr](#) • Key Utility: Physical-to-digital document processing.

13. 👥 Friends

Path: /friends

The social collaboration hub. Features live friend lists, real-time 1-on-1 direct messaging, voice notes, study group chat rooms, and peer-to-peer WebRTC video study calls.

Icon: [Users](#) • Backend Service: [/api/friends](#) , [/api/groups](#) & Socket.IO • Key Utility: Peer study, real-time chat, and video calls.

14. ⚙️ Settings

Path: /settings

Account configuration center. Allows students to edit personal details, select target exam focus (JEE, NEET, UPSC, etc.), customize profile avatars, and toggle dark mode.

Icon: `Settings` • Backend Service: `/api/profile` • Key Utility: User profile & preference management.

3. Top Navbar Controls & Interactive Utilities

Navbar Utility	Visual Element	Functional Mechanism & Code Logic
Global Search Bar	<code><Search /></code> input field	Typing a search term automatically routes the user to <code>/youtube?q=<query></code> , allowing seamless cross-platform video query dispatch. Automatically clears when navigating to non-video routes.
Theme Toggle Engine	<code><Moon /></code> / <code><Sun /></code> button	Switches application theme between Light and Dark mode using <code>ThemeContext</code> . Persists user preference in <code>localStorage</code> and HTML root class.
Notification Trigger	<code><Bell /></code> icon with indicator dot	Displays active platform notifications, study session reminders, task deadlines, and friend request alerts with an amber status dot.
User Profile Dropdown	Avatar Circle & Chevron	Opens an animated popover showing: - Student Full Name & Email Address - Target Exam Badge (🎯 JEE, NEET, UPSC, GATE, etc.) - Quick Link: My Profile - Quick Link: Settings - Action: Sign Out (clears JWT token & redirects to login)

4. Sidebar Controls & Responsiveness Mechanics

Collapsible Sidebar Mechanics (256px ↔ 68px)

Clicking the `<ChevronLeft />` or `<ChevronRight />` toggle button expands or collapses the sidebar width dynamically using Framer Motion animation. When collapsed, text labels hide smoothly, and floating hover tooltips popovers appear when hovering over icons.

Mobile Hamburger & Overlay Menu

On mobile screens (screen width < 1024px), the sidebar transitions into a slide-out drawer controlled by a fixed top-left hamburger button (`<Menu />` / `<X />`) and a semi-transparent backdrop overlay.